

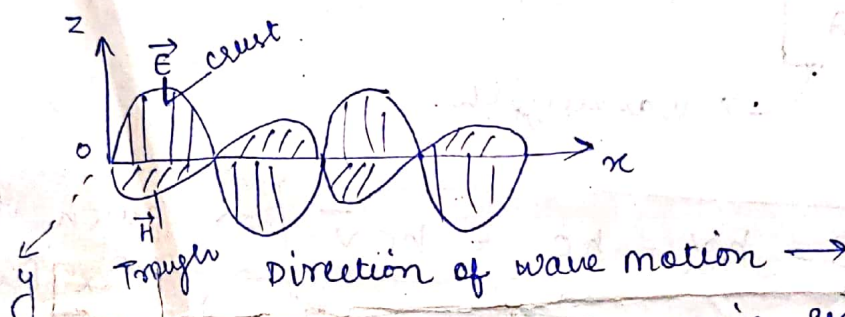
Spectroscopy

Ques.

Spectroscopy is defined as the analysis of the results coming out of interaction b/w electromagnetic radiations and matter.

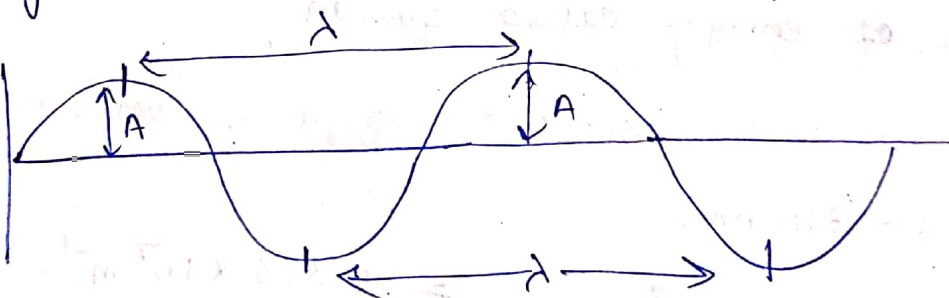
* What are Electromagnetic Radiations?

⇒ Radiations which are made up of two mutually perpendicular oscillating electric & magnetic fields & also $\perp$ to direction of propagation.



Important characteristics of spectroscopic signals

* wavelength (λ) $\Rightarrow$ Distance b/w 2 crests & 2 troughs



$$1 \text{ \AA} = 10^{-10} \text{ m}$$

$$1 \text{ mm} = 10^{-3} \text{ m}$$

$$1 \text{ \mu m} = 10^{-6} \text{ m}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

$$1 \text{ pm} = 10^{-12} \text{ m}$$

* Frequency (ν) - no. of waves passing through given point in unit time.

It is measured as cycles per second (cps) called Hertz.

$$1 \text{ kHz} = 10^3 \text{ Hz} \quad 1 \text{ MHz} = 10^6 \text{ Hz}$$

* Wavenumber ($\bar{\nu}$) = $\frac{1}{\lambda}$

* $\boxed{\nu = \frac{c}{\lambda}}$ speed of light ($3 \times 10^8 \text{ ms}^{-1}$)

frequency $\swarrow$ $\searrow$ wavelength

* Energy $\boxed{E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}}$ $\Rightarrow$ given by max. Planck.

Energy in the form of electromagnetic radiation comes discrete units called photons. Each photons contain amount of energy called quanta

Ques Convert 300 nm into frequency, waveno & energy.

$$\lambda = 300 \text{ nm}$$

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{300 \times 10^{-9} \text{ m}} = 0.333 \times 10^7 \text{ m}^{-1}$$

$$= 0.333 \times 10^7 \times 10^{-2}$$

$$\nu = \frac{c}{\lambda} = c\bar{\nu} = 3 \times 10^8 \times 33333.3 \text{ cm}^{-1}$$

$$= \boxed{9.99 \times 10^{14} \text{ s}^{-1}}$$

$$E = h\nu = 6.626 \times 10^{-34} \text{ Js} \times 9.99 \times 10^{14} \text{ s}^{-1}$$

$$= \boxed{6.25 \times 10^{-19} \text{ J}}$$

8. Dissociation energy of hydrogen molecule is $104 \text{ kcal mol}^{-1}$ express in terms of wavelength, wave no & frequency. 2

$$\nu = \frac{E}{h} = \frac{104 \times 10^3 \text{ cal mol}^{-1} \times 4.184 \text{ J cal}^{-1}}{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.626 \times 10^{-34} \text{ Js}}$$

$$= \boxed{10.91 \times 10^{14} \text{ s}^{-1}}$$

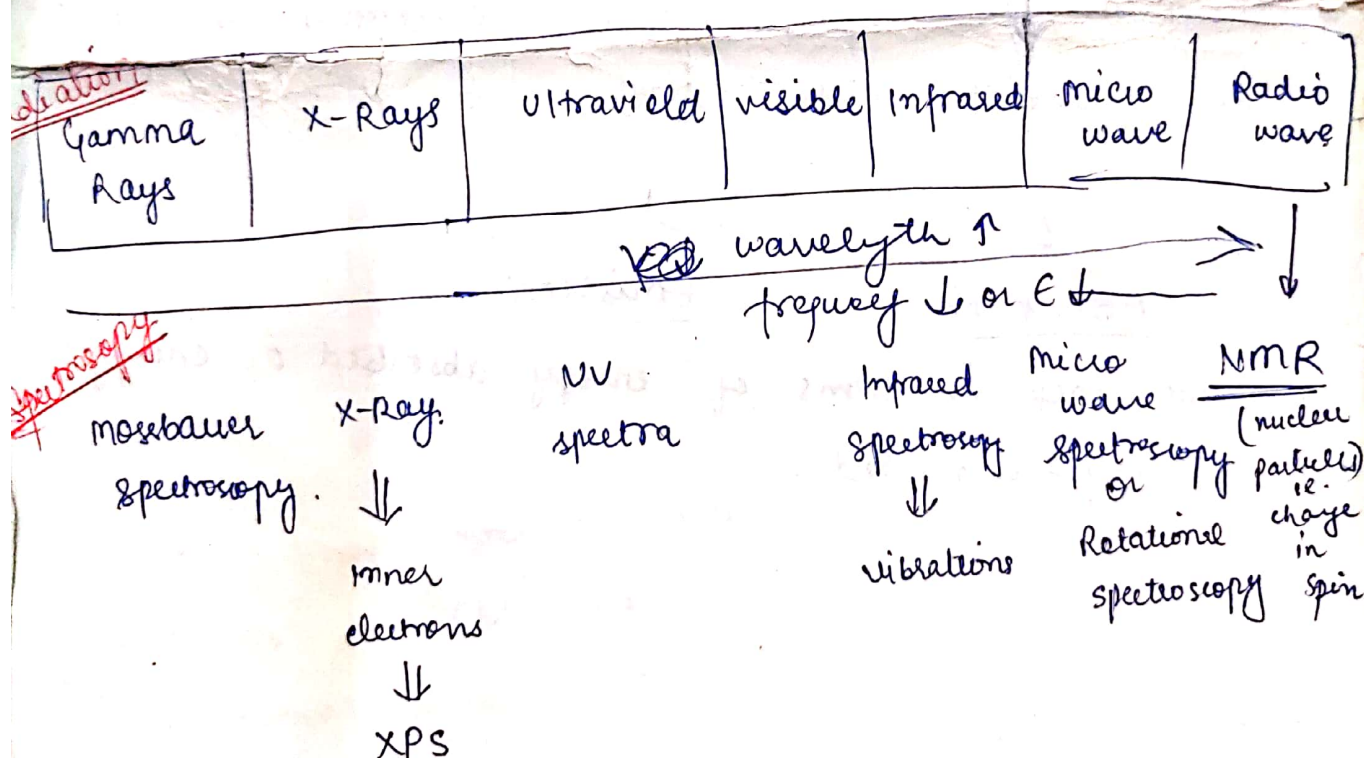
$$\bar{\nu} = \frac{\nu}{c} = \frac{10.91 \times 10^{14} \text{ s}^{-1}}{3 \times 10^{10} \text{ cm s}^{-1}} = \boxed{36362.7 \text{ cm}^{-1}}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{36362.7 \text{ cm}^{-1}} = 2.75 \times 10^{-5} \text{ cm}$$

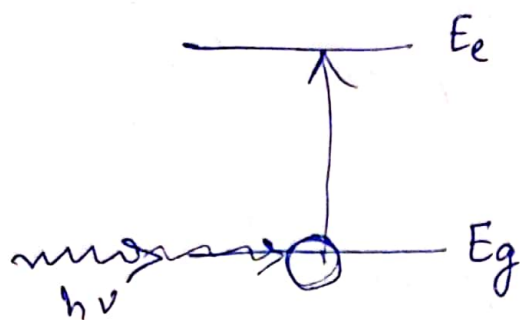
$$= 2.75 \times 10^{-5} \times 10^{-2}$$

$$= 275 \times 10^{-9} \text{ m} = \boxed{275 \text{ nm}}$$

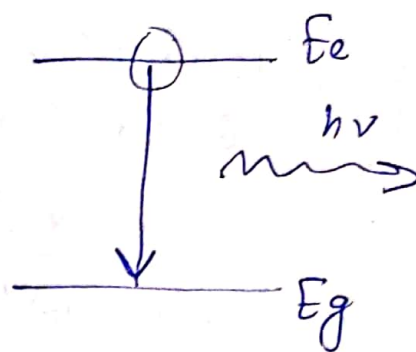
Electromagnetic Spectrum



Origin of spectrum



Absorption



Emission

$$\Delta E = E_e - E_g$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$\Delta E = h\nu$$

$$= \frac{hc}{\lambda}$$

$$= hc\bar{\nu}$$

Energy is inversely prop. to λ .

There are 2 ways in which electromagnetic waves can interact with matter.

Absorption

Emission

So we talk in terms of energy absorbed or energy emitted / transmitted.

Usefulness of Spectroscopic Techniques

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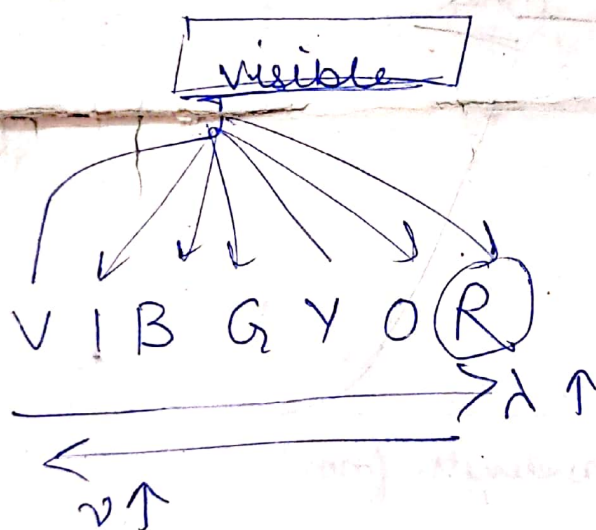
is used for determining the internal structure of organic compounds.

- Requires small amount of sample
- It takes less time
- It is very economical in long run.
ie. it is cost effective in long run.

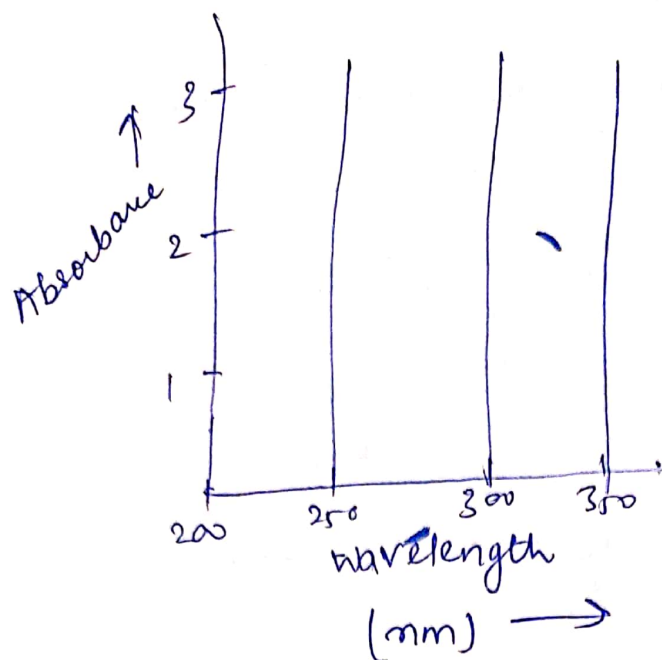
UV-visible Spectroscopy

Ultraviolet (UV) region $\Rightarrow$ 100-350 nm

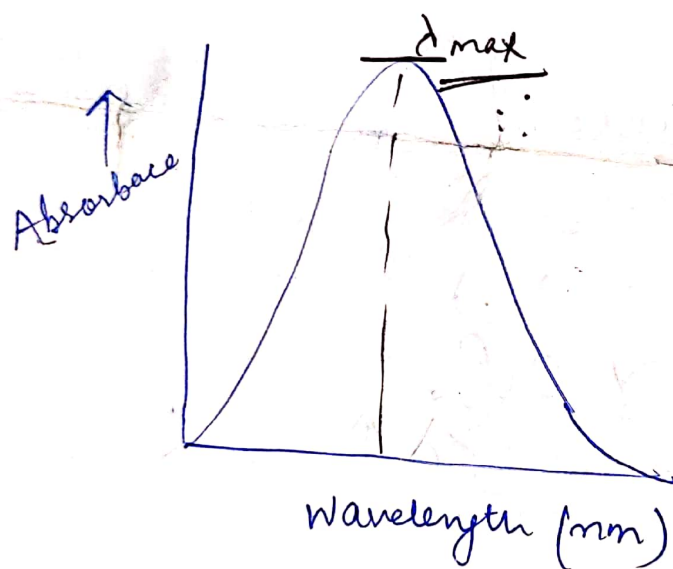
visible (VIS) " $\Rightarrow$ 350-700 nm



- * When sample interacts with EM radiations, spectrum is obtained
- * spectrum can be plotted b/w absorbed radiation as a function of frequency or wavelength.



we don't obtain such type of peak. rather we obtain a broad peak.

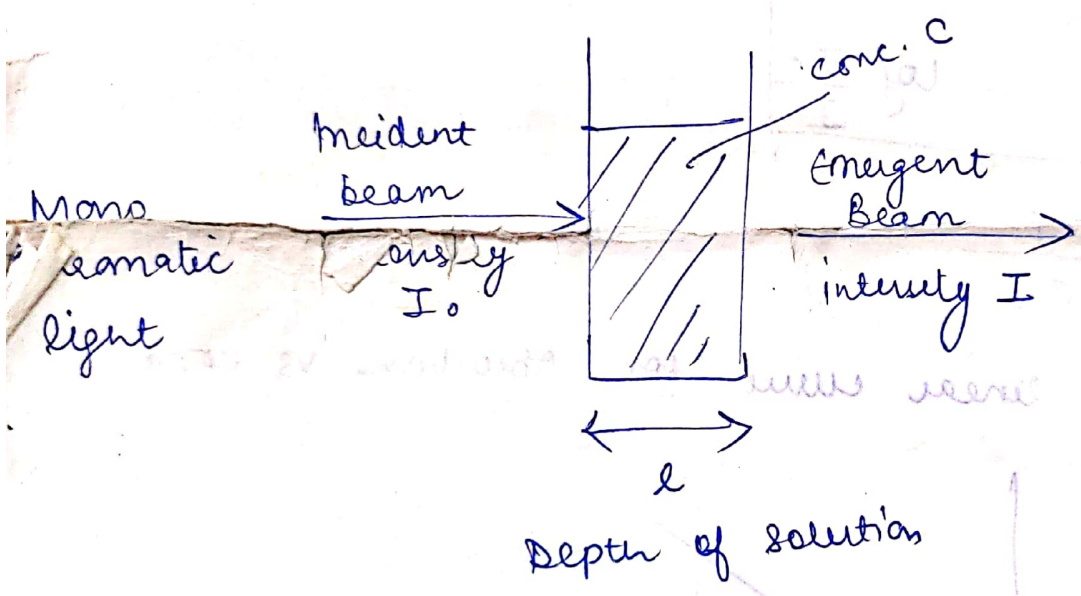


Bee's Lambert's Law

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is relationship b/w Absorbance & length & conc.
When monochromatic light of initial intensity I_0 passes through a solution in transparent vessel, some of the light is absorbed so that intensity of the transmitted light I is less than I_0 .

→ There is some loss of light intensity from scattering by particles in solution & reflection but mainly from absorption of solution.



Absorbance of the solution is directly prop. to the length of the cell, l & also to the concentration of the solution C .

$$A \propto l$$

$$A \propto C$$

$$A \propto lC \Rightarrow$$

$A = \epsilon lC$

ϵ - molar extinction coeff.

CS2008

for experimental purpose, we use Transmittance.

Ratio of intensities is known as Transmittance (T)

& usually expressed as percentage.

$$\%T = \frac{I}{I_0} \times 100$$

$$\log \%T = \left(\log \frac{I}{I_0} \right) + \log 10^2$$

$$A = -\log T$$

$$\log \%T = -A + \log 100$$

$$A = -\log \frac{I}{I_0} = \log \frac{I_0}{I}$$

$$A = 2 - \log \%T$$

$$A = \epsilon c l = \log \frac{I_0}{I}$$

$$\text{unit } \epsilon = \frac{A}{cl}$$

we expect a linear curve for Absorbance vs conc.



But sometimes, there are

certain deviations to

Bee's - Lambert's law

$$A = \epsilon c l$$

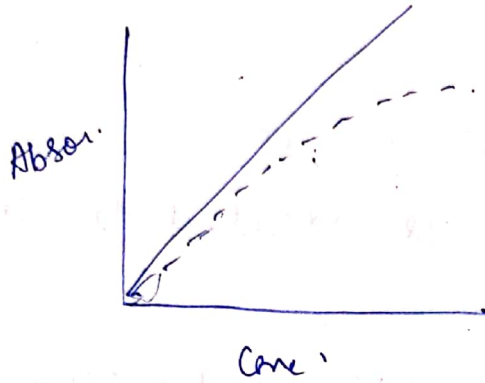
Limitation of Beer's Lambert's Law

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1) Chemical Deviations (such as association or dissociation)

Real Deviations (due to high concentrations)

Instrumental Deviations (due to polychromatic light)

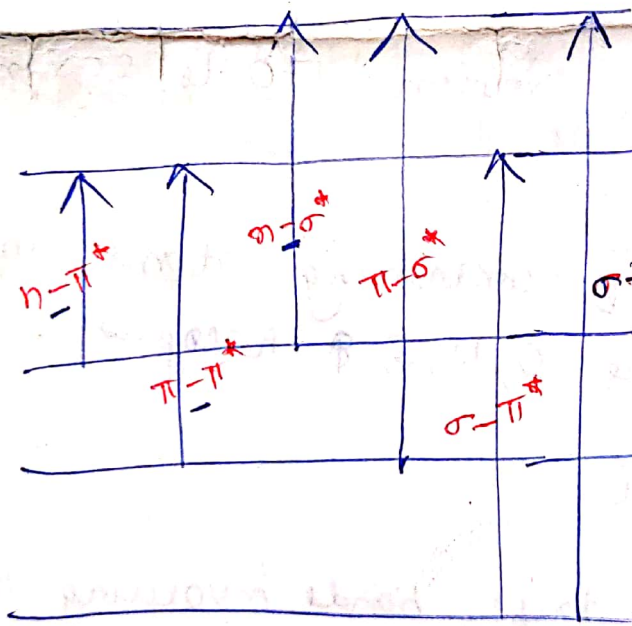


monochromatic
↓
light having single wavelength
↓
e.g. blue or red or any coloured

Polychromatic
↓
broad range of wavelengths
ex- white light coz it is comb of diff. λ

Electronic Transitions

Energy ↑



σ^* (Anti-Bonding)

π^* (Anti-Bonding)

$n \rightarrow \sigma^*$
 n (non-bonding)

π (Bonding)

σ (Bonding)

$\sigma \rightarrow \sigma^*$
 $n \rightarrow \sigma^* \Rightarrow$

1. $\sigma \rightarrow \sigma^*$ Transition

- * σ electron from orbital is excited to corresponding anti-bonding orbital σ^* .
- Energy required is very large for this transition.
- Ex - CH_4 methane has C-H bond only & can undergo $\sigma \rightarrow \sigma^*$ transition & show absorbance maxime at 125 nm.

* 2. $\pi \rightarrow \pi^*$

- π electron from orbital is excited to corresponding anti bonding π^* orbital.
- Compounds containing multiple bonds like alkenes, alkynes, carbonyls, nitriles, Aromatic compounds etc.
- Alkenes absorb in region 170 to 205 nm.

3. $n \rightarrow \sigma^*$

- Saturated compound containing atoms with lone pair of e's like O, N, S & halogens.
- require less energy.

* 4. $n \rightarrow \pi^*$

compounds containing double bonds involving hetero atoms
($\text{C}=\text{O}$, $\text{C}\equiv\text{N}$, $\text{N}=\text{O}$)

- require min. energy.

5. $\sigma \rightarrow \pi^*$

6. $\pi \rightarrow \sigma^*$

These electronic transitions are forbidden. trans. & are only theoretically possible.

only $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$ electronic transitions show absorption in the UV region (200 - 700 nm) 6

Some Terms describing UV absorption

1. Chromophore — ~~the~~ part of molecule responsible for imparting colour.

These are functional groups containing multiple bonds capable of absorbing radiations above 200 nm due to $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$

e.g. $N=O$, $C=O$, $C=N$, $C \equiv N$, $C=C$, $C=S$ etc.

2. Auxochrome — Substituents with unshared pair of electrons like OH, NH, SH that don't absorb radiation in UV but when attached to chromophore moves the absorption max. to longer λ .

Ex — Benzene $\lambda_{max} = 255 \text{ nm}$



Phenol $\lambda_{max} = 270 \text{ nm}$



Aniline $\lambda_{max} = 280 \text{ nm}$



Absorption & Intensity Shifts

1. Bathochromic shift (Red shift)

→ when absorption maxima (λ_{max}) of compd shift to longer wavelength.

→ This effect is due to presence of an auxochrome or by change of solvent.

2. Hypsochromic shift (Blue shift)

→ shifts to shorter wavelength

→ This effect is due to presence of a group causes removal of conjugation or by change of solvent.

• Hyperchromic shift — increase in Absorption intensity (ϵ) of compounds.

— It can be done by introduction of auxochrome.

Hypochoemic shift — decrease in ϵ of band.

